

Romina Soledad Molina, Ph.D.

Trieste, Italy

✉ rominasol.molina@gmail.com • ID 0000-0001-7688-6248 • 🌐 RomiSolMolina

Personal Information

Citizenship: Argentine / Italian

Birth Place: San Luis, Argentina

Education

Università degli Studi di Trieste and Universidad Nacional de San Luis <i>PhD in Industrial and Information Engineering (Italy) / Computer Science (Argentina)</i> Thesis: "SoC-based FPGA architecture for image analysis and other highly demanding applications"	Italy / Argentina 2019–2023
Universidad Nacional de San Luis <i>Master in Computer Science</i> Thesis: "Recuperación de Imágenes sobre Plataformas de Sistemas de Cómputo de Alta Productividad" (Image Retrieval on High Productivity Computing Systems Platforms)	Argentina 2014–2017
Universidad Nacional de San Luis <i>Bachelor in Electronic Engineering</i>	Argentina 2003–2010

Affiliations

2019–2026: The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy.

2019–2023: Università degli Studi di Trieste (UNITS), Italy.

2020–2024: Associazione Società Italiana di Elettronica (SIE), Italy.

2020–2023: Istituto Nazionale di Fisica Nucleare (INFN), Italy.

2021–2022: Consiglio Nazionale delle Ricerche (CNR) -ISTI, Pisa, Italy.

2012–2022: Universidad Nacional de San Luis (UNSL), Argentina.

Professional Experience

Current Position: Independent Researcher & Founder, KaleidoForge, Trieste, Italy.

- Design and delivery of hands-on training courses on neural network compression (pruning, quantization-aware training, knowledge distillation, and low-rank factorization) for edge deployment; delivered to ~70 participants across Argentina, Italy, Spain, and online (2025–2026).
- Development of educational materials for the AMD University Program on Machine Learning (ML) and FPGA-based deployment (collaboration through ICTP).
- Development of *KalEdge*: an end-to-end platform automating the compression–HLS–FPGA deployment pipeline, used as the practical framework in the above courses.
- IAEA consultancy: ML-based RIID system for nuclear instrumentation deployed on Arduino Uno Q.
- Research on FPGA accelerators for ultra-low-power edge inference.

MLab-STI/ICTP

Short-Term Visitor

Trieste, Italy

Feb–Apr 2026

- Research visit on machine learning for edge AI and FPGA-based scientific instrumentation.
- Contribution to AMD University Program documentation on ML and FPGA deployment.

MLab-STI/ICTP

TRIL Fellow (Training and Research in Italian Laboratories)

Trieste, Italy

Jul–Nov 2025

- Research fellowship on machine learning for edge AI and FPGA-based deployment.
- Initiated development of educational materials for the AMD University Program on ML and FPGA deployment.

MLab-STI/ICTP

Trieste, Italy

Postdoctoral researcher on Machine Learning for Advanced Scientific Instrumentation

2023-2025

- Research, development, and implementation of ML algorithms for advanced scientific instrumentation, targeting different devices, such as FPGA/SoC, microcontrollers, and CPU.
- AI-driven research projects in interdisciplinary fields.

Università degli Studi di Trieste – Universidad Nacional de San Luis – ICTP

Italy/Argentina

Ph.D. Student

2019–2023

- Research in model compression of machine learning models and cost estimation models.
- Acceleration of machine learning models in the context of precision agriculture, information retrieval, image processing, and particle physics.
- Supervised bachelor and master thesis projects focused on machine learning.
- ICTP instructor.

Universidad Nacional de San Luis

Argentina

Researcher and Lecturer

2012–2019

- Assistant professor: digital signal processing and digital control (Electronic Engineer degree).
- Lecturer in post-graduate courses (master and Ph.D. levels).
- Research in artificial vision and control systems.
- Supervised thesis projects focusing on machine learning and image processing.

Research/Visiting Periods in Other Centers

2024: University of Novi Sad, Serbia – Horizon 2020 research and innovation staff exchange grant.

2024: Thapar Institute of Engineering and Technology, India - Bilateral project Italy-India.

2023: International Atomic Energy Agency (IAEA) – NSIL, Vienna, Austria.

2022: NECSTLab: Advanced projects in computing systems - DEIB, Politecnico di Milano, Italy.

2021: ISTI – CNR, Pisa, Italy

2021–2023: Scientific contractual agreement, AI and HPC research on SoC-FPGA devices, ICTP, Trieste, Italy.

2020–2021: Visiting Scientist, Multidisciplinary laboratory (MLab), ICTP, Trieste, Italy.

2019–2020: Visiting Scientist, Multidisciplinary laboratory (MLab), ICTP, Trieste, Italy.

2018: Research period at Universidad de Castilla-La Mancha, Ciudad Real, Spain.

2015: Research period at Universidad de Castilla-La Mancha, Ciudad Real, Spain.

Research Experience in Specific Projects

2024–2025: REMARKABLE: Rural Environmental Monitoring via ultra wide-ARea networkS And distriButed federated Learning - Horizon 2020 research and innovation staff exchange.

2022–2024: Manageable AI for Scene Understanding Tasks from Low-Light Noisy Images, Italy and India.

2017–2018: HAMLET: Hardware Acceleration of Machine Learning Tasks, Italy and Argentina.

2015–2016: Similarity Search in the Big Data Era, Argentina.

2012–2019: Artificial Vision and Digital Control, Argentina.

Selected scholarships/grant

2024: Receiver of "Subsidio profesor visitante", Fundación Williams, Argentina [Presented by Luciana De Micco. Granted 2024, Executed 2025].

2024: IAEA scholarship, Muon Tomography Workshop, France.

2019–2023: PhD scholarship, POR FVG – Fondo Sociale Europeo, Italy.

2017: ICTP scholarship, FPGA for Nuclear Instrumentation, Trieste, Italy.

Teaching and Supervision

2012–2022: Assistant professor: Digital Signal Processing and Digital Control, Electronic Engineer, Universidad Nacional de San Luis, Argentina

- Responsible for practical exercises, simulations, and laboratories in digital signal processing and digital control.

2017–Present: Lecturer in postgraduate and specialized courses

- *From algorithm to hardware: machine learning in embedded systems* (2026). Specialized course (organized and taught), Universidad de Castilla-La Mancha, Ciudad Real, Spain.
- *Applied ML on FPGA: An End-to-End Perspective* (2026). Technical workshop (organized and taught), Online (first cohort). Two sessions × 3h.
- *From algorithm to hardware: machine learning in embedded systems* (2026). Specialized course (organized and taught), Instituto Balseiro, San Carlos de Bariloche, Argentina.
- *Inteligencia artificial en sistemas embebidos* (2025) - Specialized course (organized and taught), Online (Peru cohort). 8h.
- *From algorithm to hardware: machine learning in embedded systems* (2025). Postgraduate course (co-organized and taught), Universidad Nacional de Mar del Plata, Argentina.
- *Procesamiento digital de señales en sistemas embebidos* (2018/2019) – UNSL, Argentina.
- *Diseño avanzado de sistemas embebidos en lógica programable: Zynq APSoC, Vivado-HLS y SDSoC* (2018) – UNSL, Argentina.
- *Análisis de procesamiento paralelo y distribuido sobre datos masivos* (2018) – UNSL, Argentina.
- *Programación avanzada de Sistemas Embebidos en RTOS y Linux Embebido* (2017) – UNSL, Argentina.

2019–2026: Lecturer and laboratory instructor in ICTP workshops and schools

- School on Applied AI for Sustainable Development | smr4110. Trieste, Italy. March 2026.
- Joint ICTP-IAEA School on Detector Signal Processing and Machine Learning for Scientific Instrumentation and Reconfigurable Computing | smr4110. Trieste, Italy. October/November 2025.
- 1st Mesoamerican Workshop on Reconfigurable X-ray and Scientific Instrumentation for Cultural Heritage | smr4078. Antigua, Guatemala. June 2025.
- Workshop on Fully-Programmable Systems-on-Chip for Scientific Applications | smr 3983. Doha, Qatar. October 2024.
- Joint ICTP-IAEA School on Systems-on-Chip Based on FPGA for Scientific Instrumentation and Reconfigurable Computing | smr389. Trieste, Italy. November/December 2023.
- Joint ICTP-IAEA School on FPGA-based SoC and its Applications to Nuclear and Scientific Instrumentation | smr3765. Trieste, Italy. November/December 2022.
- Joint ICTP, SAIFR and UNESP School on Systems-on-Chip, Embedded Microcontrollers and their Applications in Research and Industry | smr3557. On-line. October 2021.
- Joint ICTP-IAEA School on FPGA-based SoC and its Applications for Nuclear and Related Instrumentation | smr3562. Trieste, Italy. January/February 2021.
- ICTP School and Conference on Fully-Programmable Systems-On-Chip for Scientific Instrumentation | smr3332. Guwahati, India. November 2019.
- ICTP Advanced Workshop on Modern FPGA Based Technology for Scientific Computing. Trieste, Italy. May 2019.
- ICTP Advanced Workshop on FPGA-based Systems-On-Chip for Scientific Instrumentation and Reconfigurable Computing | smr3332. Trieste, Italy. November/December 2018.

2018–2026: Oral Presentations, invited lectures, and tutorials

- Fast Machine Learning for Science Conference 2026, San Diego, California, EE.UU. (2026).
Tutorial (to be held): KalEdge: Bridging ML models and Edge-AI.
- 3rd FPGA Developers Forum, CERN, Geneva, Switzerland (2026).
Talk: KalEdge-Lite: Hardware-Aware ML-to-FPGA Deployment with Automated hls4ml Integration.
- Technical Seminar, Celestica Inc. (2026).
Topic: Building Next Gen Edge AI with FPGA.
- Summer School: IoT for Eco-Friendly Tourism, Valencia, Spain (2025).
Topic: Machine learning and model compression for FPGA-based Edge IoT devices.

- Presentation: Workshop on Seismology and Artificial Intelligence, Frankfurt Institute for Advanced Studies, Frankfurt, Germany (2023).
Topic: Seismic event detection at Copahue volcano using digital signal processing and machine learning: towards edge implementation.
- DEIB – NECSTLab, Politecnico di Milano, Milan, Italy (2022).
Topic: SoC-FPGA architectures for machine learning applications.
- Workshop: CAHEP2020 – Central American Meeting on High Energy Physics, Cosmology, and High Energy Astrophysics, Guatemala (November 2020).
Topic: “FPGA para la aceleración de algoritmos de Machine Learning.”
- Universidad de Castilla-La Mancha (UCLM), Spain (2018).
Seminar: “Digital Signal Processing” - Master’s in Computer Engineering (Ingeniería Informática).

2014-2023 - Supervision: Supervised multiple thesis projects in image processing and AI, focused on interdisciplinary fields.

- Selected thesis
 - *FPGA-Oriented Machine Learning for Pulse Arrival Time Detection in Radiation Detectors*. Student: Lucas Starita. Supervisor: Luciana de Micco – Co-supervisor: Romina Molina. Year: 2026/2027. Ongoing. [National University of Mar del Plata, Argentina - In collaboration with Laboratory for Ion Beam Interactions, Ruđer Bošković Institute, Croatia].]
 - *Machine learning techniques for moth classification in precision agriculture*. Student: Valentina Carrer. Supervisor: Giovanni Ramponi – Co-supervisor: Romina Molina. Year: 2022. [Master in Computer and Electronic Engineering (Laurea magistrale, UniTS, Trieste, Italy).]
 - *Herramienta basada en procesamiento digital de señales y aprendizaje automático para la clasificación de trazas sísmicas volcánicas*. Student: Yair Mauad Sosa. Supervisor: Romina Molina. Co-supervisor: Alejandro Nuñez Manquez. Year: 2023 [Bachelor Electronic Engineer with Digital System Orientation (UNSL, San Luis, Argentina).]
 - *Detección de plagas en cultivos frutales a través de procesamiento de imagen y aprendizaje automático mediante trampas IoT*. Student: Agustina Suarez. Supervisor: Romina Molina. Co-supervisor: Ricardo Petrino. Year: 2021. [Bachelor Electronic Engineer with Digital System Orientation (UNSL, San Luis, Argentina).]
 - *Equipo de medición de velocidades de semillas en descargas de silos con visión*. Student: Valeria Gonzalez. Supervisor: Ing. Romina Molina – Co-supervisor: Jéscica Benito. Year: 2019. [Bachelor Electronic Engineer with Digital System Orientation (UNSL, San Luis, Argentina).]

Awards and Recognitions

2022: Industry Impact Award, ECIR2022, Norway.

List of Publications

Book Chapters:

- Gil Costa, **Molina, R. S.**, et al., *Hardware Acceleration of CBIR System with FPGA-Based Platform*, IGI, 2016.

Journal Papers:

- **Molina, R. S.**, et al., *A Reconfigurable Spiking Neural Network on FPGA for Radiation Pulse Discrimination*. Under review. Elsevier Nuclear Engineering and Technology. 2026.
- Moraga, A., Ballina, M., **Molina, R. S.**, et al., *A reconfigurable edge IoT framework for machine learning-based precision agriculture*. Under review. Elsevier Agricultural Systems. 2026.
- Alziyad, M., Hassan Al-Zahrani, M., **Molina, R. S.**, et al., *SNeuron: Learnable Structural Neuron Gating for Energy-Efficient Embedded AI*. Under review. IEEE Embedded System Letters Journal. 2026.
- Bhattacharya, J., **Molina, R. S.**, et al., *A Feature Distillation Network to Enable Object Detection on an FPGA Platform in Poor Visibility Conditions*, MDPI Electronics. 2026.
- **Molina, R. S.**, et al., *Physics-Guided Neural Network for Pile-Up Rejection on SoC-FPGAs*, IEEE TNS. 2025
- **Molina, R. S.**, et al., *Advanced Neutron Discrimination with Diamond Detectors Using Machine Learning*.

- Advanced Neutron Discrimination with Diamond Detectors Using Machine Learning. IEEE transactions on nuclear science.
- Garcia, L. G., Crespo, M. L., Cicuttin, A., Morales, I., **Molina, R. S.**, et al. *MA-XRF Scanner: Implementation of Elemental Maps Reconstruction Algorithm Based on Single Photon Detection for Cultural Heritage Studies*, X-Ray Spectrometry, Wiley. 2025.
 - Ballina, M. G., Crespo, M.L., Carrato, S., **Molina, R. S.**, et al., *A method for accurate and precise pulse arrival time estimation: A case study on high-energy particle detectors*, IEEE Transactions on Nuclear Science. 2025. doi: 10.1109/TNS.2025.3557623.
 - Morales, I., **Molina, R. S.**, et al., *Embedded real-time high event rate gamma/neutron discrimination based on machine learning with SiPM CLYC detectors*, IEEE Transactions on Nuclear Science. 2024.
 - Sosa, Y. M., **Molina, R. S.**, et al., *Seismic Event Detection in the Copahue Volcano Based on Machine Learning: Towards an On-the-Edge Implementation*, MDPI Electronics. 2024.
 - **Molina, R. S.**, et al., *An End-to-End Workflow to Efficiently Compress and Deploy DNN Classifiers On SoC/FPGA*, IEEE Embedded Systems Letters. 2023.
 - **Molina, R. S.**, et al., *High-Level Synthesis Hardware Design for FPGA-based Accelerators: Models, Methodologies, and Frameworks*, IEEE Access. 2022.
 - Cicuttin, A., Morales, I. R., Crespo, M. L., Carrato, S., García, L. G., **Molina, R. S.**, et al., *A Simplified Correlation Index for Fast Real-Time Pulse Shape Recognition*, MDPI Sensors. 2022.
 - **Molina, R. S.**, et al., *Efficient traversal of decision tree ensembles with FPGAs*, Journal of Parallel and Distributed Computing, Elsevier. 2021.
 - Garcia, L. G., **Molina, R. S.**, et al., *Muon–Electron Pulse Shape Discrimination for Water Cherenkov Detectors Based on FPGA/SoC*, MDPI Electronics. 2021.
 - Marsi, S., Bhattacharya, J., **Molina, R. S.**, et al., *A Non-Linear Convolution Network for Image Processing*, MDPI Electronics. 2021.
 - Guzzi, F., De Bortoli, L., **Molina, R. S.**, et al., *Distillation of an End-to-End Oracle for Face Verification and Recognition Sensors*, MDPI Sensors. 2021.
 - **Molina, R. S.**, et al., *Heterogeneous SoC-based acceleration of MPEG-7 compliance image retrieval process*, Journal of Real-Time Image Processing. 2018.

Conference Papers:

- **Molina, R. S.**, *KalEdge: Bridging Neural Network Compression and FPGA Deployment in a Unified Workflow*. Accepted. International Conference on Applications in Electronics Pervading Industry, Environment and Society 2026.
- Ballina, M., **Molina, R. S.**, et al., *K'awiil: A Framework for Profiling Heterogeneous MPSoC-FPGA Systems for Machine Learning Inference*. Accepted. International Conference on Applications in Electronics Pervading Industry, Environment and Society 2026.
- Anzorena, S., **Molina, R. S.**, et al., *Hardware-Aware Deep Learning for Edge-Based Sleep Apnea Detection from Single-Lead ECG*. Under review. IEEE ARGENCON 2026.
- **Molina, R. S.**, et al., *Holistic Edge AI: From ML Models to Hardware-Aware Deployment*, IEEE EDUCON 2026.
- **Molina, R. S.**, et al., *Efficient Split Learning LSTM Models for FPGA-based Edge IoT Devices*, IEEE International Conference on Machine Learning for Communication and Networking. 2025.
- Ballina, M., **Molina, R. S.**, et al., *HyperFPGA: Enhancing Education With Remote Laboratory Access for Heterogeneous Computing on MPSoC-FPGA Technologies*, IEEE EDUCON. 2025.
- García Ordóñez, L. G., Crespo, M. L., Cicuttin, A., Morales, I., **Molina, R. S.**, et al., *MA-XRF Scanner: Implementation of Image Reconstruction Algorithms Based on Single Photon Detection for Cultural Heritage Studies*, EXRS2024 conference. 2024.
- **Molina, R. S.**, et al., *ML-based classifier for precision agriculture on embedded systems*, International Conference on Applications in Electronics Pervading Industry, Environment and Society. 2022.
- Florian Samayoa, W., Valinoti, B., **Molina, R. S.**, et al., *Diagnostic analytics for pixelated particle detectors: A case study*, International Conference on Applications in Electronics Pervading Industry, Environment and Society. 2022.
- Gil-Costa, V., Loor, F., **Molina, R. S.**, et al., *Energy-Efficient Ranking on FPGAs through Ensemble Model*

- Compression*, 12th Italian Information Retrieval Workshop. 2022.
- **Molina, R. S.**, et al., *Compression of NN-Based Pulse-Shape Discriminators in Front-End Electronics for Particle Detection*, International Conference on Applications in Electronics Pervading Industry, Environment and Society. 2022.
 - Gil-Costa, V., Loor, F., **Molina, R. S.**, et al., *Ensemble Model Compression for Fast and Energy-Efficient Ranking on FPGAs*, European Conference on Information Retrieval, Springer. 2022.
 - Suárez, A., **Molina, R. S.**, et al., *Pest detection and classification to reduce pesticide use in fruit crops based on deep neural networks and image processing*, IEEE XIX Workshop on Information Processing and Control (RPIC). 2021.
 - García Ordóñez, L. G., **Molina, R. S.**, et al., *Pulse shape Discrimination for Online Data Acquisition in Water Cherenkov Detectors Based on FPGA/SoC*, 37th International Cosmic Ray Conference. 2021.
 - **Molina, R. S.**, et al., *Implementation of Particle Image Velocimetry for Silo Discharge and Food Industry Seeds*, Applications in Electronics Pervading Industry, Environment and Society. Lecture Notes in Electrical Engineering. 2021.
 - Guillermo, G. L., Crespo, M. L., Carrato, S., Cicuttin, A., Oswaldo, F. W., **Molina, R. S.**, et al., *High Voltage Isolated Bidirectional Network Interface for SoC-FPGA Based Devices: A Case Study: Application to Micro-pattern Gaseous Detectors*, Applications in Electronics Pervading Industry, Environment and Society. Lecture Notes in Electrical Engineering. 2021.
 - **Molina, R. S.**, et al., *Implementation of Search Process for a Content-Based Image Retrieval Application on System on Chip*, X Southern Conference on Programmable Logic. 2019.
 - Ariza, C. R., **Molina, R. S.**, et al., *Eye Tracker Acceleration in Reconfigurable Hybrid Systems*, IEEE Biennial Congress of Argentina (ARGENCON). 2018.
 - Luis Britos, Maria E. Di Gennaro, Veronica Gil Costa, Fernando Kasián, Jair Lobos, Verónica Ludueña, **Molina, R. S.**, et al., *Búsquedas en Grandes Volúmenes de Datos*, WICC 2016 - XVII Workshop de Investigadores en Ciencias de la Computación. 2016.
 - **Molina, R. S.**, et al., *Conversión de RGB a YCbCr en SystemGenerator y HLS*, XXI Congreso Argentino de Ciencias de la Computación CACIC. 2015.
 - **Molina, R. S.**, et al., *Hybrid Classification of Resistors through Image Processing*, 22nd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing. 2014.

Academic Training and Events

- Joint IAEA-ICTP Workshop on Artificial Intelligence and Machine Learning in Advancing Nuclear Engineering. Abdus Salam International Centre for Theoretical Physics (ICTP). Trieste, Italy. 2025.
- Joint IAEA–French Alternative Energies and Atomic Energy Commission Workshop on Muon Tomography: From Basic Principles to Practical Usage and Applications, Gif-sur-Yvette, France, Jun 2024.
- Workshop Seismology and Artificial Intelligence, Frankfurt Institute for Advanced Studies, Frankfurt, Germany. 2023.
- ApplePies2022: Applications in Electronics Pervading Industry, Environment and Society. Genova, Italy. 2022.
- 44th European Conference on Information Retrieval (ECIR). Stavanger, Norway. 2022.
- ApplePies2021: Applications in Electronics Pervading Industry, Environment and Society. 2021. Hybrid mode.
- SIE2021: 52nd Annual Meeting of the Associazione Società Italiana di Elettronica (SIE), Trieste, Italy. 2021.
- International Graduate School for PhD students in Electronics. Trieste, Italy. 2021.
- ApplePies2020: Applications in Electronics Pervading Industry, Environment and Society. 2020. On-line.
- 17th International Workshops on Hadron Structure and Spectroscopy (IWHSS 2020), Trieste, Italy. 2020. On-line.
- Second International Conference on Advances in Electrical, Electronic and System Engineering (ICAESE 2019), Guwahati, India. 2019.
- X Southern Programmable Logic Conference, Buenos Aires, Argentina. 2019.
- Joint ICTP-IAEA School on Zynq-7000 SoC and its Applications for Nuclear and Related Instrumentation, Abdus Salam International Centre for Theoretical Physics (ICTP). Trieste, Italy. 2017.
- VII Congreso de Microelectrónica Aplicada, San Luis. Argentina. 2016.
- 4th Embedded system school, San Luis, Argentina. 2015.

- Simposio Argentino de Sistemas Embebidos (SASE), Buenos Aires, Argentina. 2014.
- 9th Escuela Argentina de Micro-Nanoelectrónica, Tecnología y Aplicaciones (EAMTA), UTN, Mendoza, Argentina. 2014.
- 8th Congreso Argentino de Micro-Nanoelectrónica, Tecnología y Aplicaciones (CAMTA), UTN, Mendoza, Argentina. 2014.
- RPIC 2013 – XV Reunión de trabajo en Procesamiento de la Información y Control, Bariloche, Río Negro, Argentina. 2013.
- Simposio Argentino de Sistemas Embebidos (SASE), Buenos Aires, Argentina. 2013.

Conference Role and Organization

Conference Organization:

- SPL 2023 (XI Southern Programmable Logic Conference), San Luis, Argentina.
- SPL 2019 (X Southern Programmable Logic Conference), Buenos Aires, Argentina.
- VII Congreso de Microelectrónica Aplicada, San Luis, Argentina (2016).

Conference Roles:

- IEEE EDUCON 2026 - Session Chair.
- SPL2023 – Technical committee.
- LAWCC 2023 - XV Congreso de la Mujer Latinoamericana en la Computación – Program committee.
- SPL2019 – Executive committee - Publicity chair.

Outreach Activities:

- "Street Science" 2025, Trieste, Italy.
- Scientific presentation "Apprendimento Automatico Applicato (Applied Machine Learning)" for the Liceo Cotta (Verona), ICTP, Trieste, Italy. 2024.
- ESOF 2020, ICTP booth, Trieste, Italy.
- Trieste Next 2019, 2021, and 2023, ICTP booth, Trieste, Italy.

Certifications & Courses (most recent)

- Supervised Machine Learning: Regression and Classification. 2025. Coursera.
- Bootcamp Avanzado MLOps | Machine Learning Operation Hands-on. 2025. Udemy.
- Entrepreneurship for Engineers. 2025. edX.
- Linear Algebra for Machine Learning and Data Science. 2023. Coursera.
- Developing FPGA-accelerated cloud applications with SD Accel: Practice. 2020. Coursera.

Technical Skills

Programming: Python, C/C++, MATLAB, VHDL.

GUI design: Qt, PyQt, Streamlit.

AI Frameworks: TensorFlow, Keras, QKeras, TF-MOT, PyTorch, Scikit-learn.

Edge AI & HLS: hls4ml, FINN, Edge Impulse, Vitis HLS, Vivado, Xilinx SDSoc.

Hardware: FPGA, SoC, Embedded Systems, Microcontrollers.

Tools: OpenCV, Git/GitHub, Jira, VS Code.